Caleb Fernandes

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EDUCATION

University of South Florida, Genshaft Honors College — Tampa, FL
Bachelor of Science in Computer Science — Expected May 2027
Bachelor of Science in Econometrics and Quantitative Economics — **GPA: 4.0**
Minor in Geographic Information Systems

TECHNICAL EXPERTISE

- **Programming:** Python, SQL, R, C++, C, C#, VBA
- **Machine Learning:** Graph Neural Networks, Gradient-Boosted Trees, Computer Vision, LLMs, Synthetic Data Generation
- **Statistics & Econometrics:** Causal Inference, Difference-in-Differences, Event Studies, Panel Regression, Time-Series Analysis, Statistical Inference
- **Spatial & Scientific Computing:** Geographic Information Systems, PyTorch, Network Analysis, pandas, sf, terra, ggplot2

WORK EXPERIENCE

Wellington Management — June 2026 – August 2026
Software Engineer Intern — *Boston, MA*

- Reverse-engineered a legacy Order Management System spanning 30 relational database tables, mapping its hierarchical rules architecture and developing a ReactJS/SQL interface to visualize, configure, and audit trade-routing logic.
- Developed tooling for quantitative execution teams to configure routing rules and backtest changes without direct software-engineering intervention, replacing a DBR-based modification process that previously required four weeks.
- Built order audit-trail functionality to trace conditional routing decisions between automated low-touch execution and high-touch trader workflows; solution is being transitioned to Trading Technology for production integration.

Gravic Inc. — May 2024 – August 2024
Software Developer Intern — *Malvern, PA*

- Saved over $20,000 annually by developing C# and VBA libraries to replace Syncfusion dependencies in automated Excel reporting workflows.
- Built .NET libraries using ClosedXML and OpenXML to programmatically generate Excel reports through C# and VBA applications.

RESEARCH EXPERIENCE

Honors Thesis — August 2025 – Present
Undergraduate Econometrics Researcher — *University of South Florida, FL*

- **Nuclear Power Plants, Local Tax Revenue, and School Spending:** Developed an end-to-end causal inference study across 52 U.S. nuclear power plants and 103,270 school districts, constructing multi-decade plant, school-finance, geographic, and state tax-policy datasets to quantify the local fiscal effects of nuclear development.
- Designed stacked event-study analyses around plant construction and operation, comparing host school districts and counties against geographically matched controls; estimated post-construction property-tax gains reaching approximately $7,000 per pupil in host districts and $2,300 per pupil at the county level.
- Extended the analysis from tax revenue to public welfare, finding that nuclear-induced fiscal gains are followed by increased educational spending in host districts and developing state-level tax classifications to study heterogeneity in the timing and distribution of fiscal benefits.

Connected Social Artificial Intelligence Lab (CSAIL) — August 2025 – Present
Graph Machine Learning Researcher — *University of South Florida, FL*

- **Florida Blue Digital Twins:** Lead an undergraduate research team developing graph- and LLM-based methods for synthetic electronic health record (EHR) generation; developed EHR-Walk, a random-walk algorithm for generating synthetic patient data from heterogeneous EHR graphs.

- **Graph Transfer Learning Team:** Developed baseline graph transfer-learning models and theoretical analysis of computation-graph approaches to embedding differentiation; present current graph machine-learning literature at weekly research meetings.

NSF REU – James Madison University Mathematics and Statistics May 2025 – April 2026
Lead Graph Machine Learning Researcher *James Madison University, VA*

- **ReCYCLE Net:** Developed a novel feature-free, unsupervised Graph Neural Network for community detection that integrates Renewal Non-Backtracking Random Walk (RNBRW) cycle information directly into graph-attention mechanisms and a custom multi-objective learning framework.
- Evaluated ReCYCLE Net across social, citation, and biological networks up to 56,944 nodes, achieving 0.344 ONMI on overlapping protein-complex detection versus 0.243 for BigCLAM, 0.232 for GAT, and 0.140 for DMoN; first-authored the resulting paper published at the GRaM Workshop at ICLR 2026.

Dr. Ran Tao's Spatial and Transportation Lab October 2024 – April 2026
Machine Learning and Spatial Researcher *Wuhan University, China*

- **Predicting Growing Disparity in Electric Charging Station Infrastructure Growth:** Built an end-to-end geospatial machine learning pipeline across 1,048,576 National Highway System road segments and 83,507 census tracts, integrating 15 demographic, transportation, and infrastructure features to predict two-year EV charging infrastructure growth.
- Developed a LightGBM hurdle model to predict the location and magnitude of charging-station growth, achieving 0.919 AUC for growth classification and $R^2 = 0.77$ for growth magnitude; research supports an in-preparation manuscript on emerging disparities in U.S. EV charging infrastructure.
- **Evaluating Spatial Disparities in Public EV Charging Infrastructure:** Developed a Python-based spatial accessibility model to quantify county-level access to public EV charging infrastructure across the United States.

Secure & Intelligent Systems (SIS) Lab May 2024 – August 2025
Computer Vision Researcher and Data Developer *University of South Florida, FL*

- **SegGen Synthetic Data Generation:** Developed the Procedural Content Generation (PCG) implementation in Unreal Engine 5 for SegGen, an automated synthetic-data pipeline that generates heterogeneous virtual environments for multimodal semantic segmentation.
- Contributed to a published synthetic-data pipeline producing automatically aligned RGB, depth, and pixel-level ground-truth labels; acknowledged for the PCG implementation in *SegGen: An Unreal Engine 5 Pipeline for Generating Multimodal Semantic Segmentation Datasets* (Sensors, 2025).

PUBLICATIONS

1. Caleb Fernandes and Behnaz Moradi-Jamei.
ReCycle Net: Cycle-Aware, Feature-Free GNN for Community Detection.
Proceedings of GRaM: the Second Edition of the Workshop on Geometry-grounded Representation Learning and Generative Modeling, PMLR 326:184–200.

2. T. M. Junaid Bashar, Ran Tao, Caleb Fernandes, and Zhenzhi Jiao.
Evaluating Spatial Disparities in Public EV Charging Infrastructure Across the United States.
Journal of Transport Geography, Vol. 130, Article 104507, 2026. DOI: 10.1016/j.jtrangeo.2025.104507.

3. Caleb Fernandes, T. M. Junaid Bashar, Ran Tao, and Zhenzhi Jiao.
Predicting Growing Disparity in Electric Charging Station Infrastructure Growth in America.
In Preparation.

4. Alec Braynen, Abel Le, Caleb Fernandes, and Seungbae Kim.
Struct-G: Structural-Aware Pretraining for Robust Graph Transfer Learning.
Submitted: ICLR 2026.

CONFERENCES AND PRESENTATIONS

[1] C. A. Fernandes *et al.*, *ReCYCLE Net: Beyond GNNs for Large-Scale Community Detection*, poster presentation at the ICLR 2026 Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRaM), Rio de Janeiro, Brazil, April 26, 2026.

[2] C. A. Fernandes *et al.*, *ReCYCLE Net: Beyond GNNs for Large-Scale Community Detection*, poster presentation at the AMS–PME Undergraduate Student Poster Session I, Joint Mathematics Meetings 2026, Washington, DC, January 6, 2026.

[3] C. A. Fernandes *et al.*, *Evaluating Spatial Disparities in Public EV Charging Infrastructure Across the United States*, oral presentation at the Gulf Coast Undergraduate Research Symposium, Rice University, Houston, TX, October 20, 2025.

[4] S. Stoke and C. A. Fernandes *et al.*, *Deep ReCYCLE: Cycle-Aware Community Detection with RNBRW-Augmented GNNs*, poster presentation at the JMU 2025 REU Summer Symposium, James Madison University, Harrisonburg, VA, July 28, 2025.

[5] C. A. Fernandes *et al.*, *Unreal Engine 5 for Synthetic Dataset Creation*, poster presentation at the Florida Undergraduate Research Symposium, University of South Florida, Tampa, FL, February 14, 2025.

AWARDS

2026 Barry Goldwater Scholar Mar. 27, 2026

One of the oldest and most prestigious national undergraduate scholarships in the natural sciences, engineering, and mathematics in the United States

Gulf Coast Undergraduate Research Symposium (GCURS) Award in Economics Oct. 20, 2025

Presented Evaluating Spatial Disparities in Public EV Charging Infrastructure Across the United States and awarded a glass, engraved trophy for the best presentation in economics

Inducted Sigma Xi Apr. 12, 2025

Inducted into the scientific research honors society, Sigma Xi

SFI Introduction to Complexity Certificate Nov. 9, 2025

Certificate for completing the Santa Fe Institute course in Introduction to Complexity

ACTIVITIES

Private Equity Club, *President of Mathematical Finance Division* Apr. 2025 – Present

Lead the Mathematical Finance Division in studying stochastic calculus, Brownian motion, Monte Carlo simulation, and their applications to quantitative finance

Sigma Xi Scientific Research Honors Society, *Member* Apr. 2025 – Present

National scientific research honors society dedicated to promoting scientific research

Theta Tau Professional Co-ed Engineering Fraternity, *Outer Guard* Aug. 2023 – Aug. 2025

Outer Guard: in charge of organizing the recruitment of new members and maintaining public relations